Global Gold Trade Analysis Report

Gold Reserves & Trade Data Analysis: 2019-2023

1. Introduction

This report provides an in-depth look at global gold trade and reserve data from 2019 to 2023. It explores international gold movements, accumulation patterns, and trends in gold reserves, focusing on BRICS nations compared to US and EU countries.

The dataset includes detailed information from 166 countries with 1,338 records, capturing import and export data for gold in its raw or semi-manufactured forms. Gold has long been a store of value and a way to protect against economic uncertainty, making reserve patterns crucial for understanding global economic changes.

The main goals of this analysis are to:

- Compare average gold purchases per country per year between BRICS members and US/EU countries
- Examine the overall trend in gold accumulation, whether it is increasing, decreasing, or stable
- Investigate the links between gold accumulation and USD dominance
- Identify which country has increased its gold reserves the most in recent years

2. Data Preparation

The dataset, which has 1,338 records and 47 columns, contains comprehensive data on the gold trade in a number of different dimensions. The main columns utilized in this analysis are described below:

Column Name	Description

refYear	Reference year of the gold trade data (2019-2023)
reporterDesc	Name of the reporting country
reporterISO	ISO 3-letter country code of the reporter
flowCode	Trade flow indicator: M (Import) or X (Export)
flowDesc	Trade flow description: Import or Export
cmdDesc	Commodity description: Gold (including gold plated with platinum) unwrought or in semi-manufactured forms, or in powder form
netWgt	Net weight of gold in kilograms (converted to tonnes in analysis)
gold_tonnes	Calculated field: netWgt / 1000 (main analysis metric)
primaryValue	Total monetary value of the gold trade in USD
partnerDesc	Trading partner (in this dataset: World)
classificationCode	Harmonized System classification version (H2, H3, H4, H5, H6)

Dataset Summary:

• Total Records: 1,338

• Countries Covered: 166

• Time Period: 2019-2023 (Annual Data)

• Trade Flows: 696 Imports, 642 Exports

• Primary Metric: gold_tonnes (netWgt converted from kg to tonnes)

• Commodity: Gold in unwrought or semi-manufactured forms

3. Data Preprocessing & Statistical Overview

3.1 Basic Statistical Summary

The gold_tonnes column (derived from netWgt) serves as the primary metric for analyzing gold movements. Key statistics include:

Statistic	Value (Tonnes)
Count	1,064 observations
Mean	64.65 tonnes
Median	0.74 tonnes
Standard Deviation	232.28 tonnes
Minimum	0.00 tonnes
25th Percentile	0.008 tonnes
75th Percentile	14.85 tonnes
Maximum	2,687.09 tonnes

Key Insights:

1. **Extremely Skewed Distribution:** The median (0.74 tonnes) is about 87 times lower than the mean (64.65 tonnes). This shows that while the majority of gold transactions are very small (less than one tonne), the average is being greatly raised by a few very large entries.

2. **Massive Concentration at the Top:** The 75th percentile (14.85 tonnes) is almost 181 times smaller than the maximum value (2,687.09 tonnes). This implies that transactions accounting for the great majority of the dataset's total gold volume are found in the top 25% of the data.

3. **High Volatility:** The extreme variability in gold movements is indicated by the standard deviation (232.28 tonnes), which is significantly greater than the mean (64.65 tonnes). This illustrates how large-scale gold trading is concentrated among important financial centers.

3.2 Top 10 Countries by Total Gold Accumulation

Rank	Country	Avg Gold/Record	Total Gold
1	Switzerland	1,191.49 tonnes	10,723.40 tonnes
2	United Arab Emirates	1,014.94 tonnes	9,134.49 tonnes

3	United Kingdom	790.33 tonnes	7,903.31 tonnes
4	China, Hong Kong SAR	588.88 tonnes	5,888.85 tonnes
5	China	510.13 tonnes	5,101.27 tonnes
6	USA	383.88 tonnes	3,838.81 tonnes
7	India	379.61 tonnes	3,796.05 tonnes
8	Canada	295.15 tonnes	2,656.39 tonnes
9	Peru	247.29 tonnes	1,978.29 tonnes
10	Singapore	260.66 tonnes	1,824.61 tonnes

Key Insights:

1.Switzerland Dominates: With a total accumulation of 10,723 tonnes, Switzerland leads the dataset. According to statistics, it has the highest average per observation (1,191 tonnes) among the leading reporters, solidifying its position as a major hub for global gold trading.

2.UAE as Major Player: With a total of 9,134 tonnes, the United Arab Emirates comes in second. In comparison to other countries, it exhibits a highly concentrated and high-value pattern of gold movement, with an average of 1,014 tonnes per entry.

3.Asian Concentration: Four of the top ten nations—the United Arab Emirates, Hong Kong, China, India, and Singapore—are located in Asia, underscoring the region's strategic emphasis on gold accumulation and its status as a significant hub for gold trade.

3.3 Yearly Gold Accumulation Trends (2019-2023)

Year	Yearly Total	Yearly Average
2019	12,076.30 tonnes	55.40 tonnes
2020	10,733.96 tonnes	45.29 tonnes
2021	12,233.24 tonnes	53.89 tonnes
2022	15,839.57 tonnes	78.80 tonnes
2023	17,901.08 tonnes	98.90 tonnes

Key Insights:

1. Strong Growth Trajectory: Over the course of five years, the global gold accumulation in this dataset increased by 48.2%, from 12,076 tonnes in 2019 to 17,901 tonnes in 2023. This indicates a notable increase in reserve accumulation and gold trading.

2. Accelerating Trend: Between 2021 and 2023, there was a significant increase. In particular, the average amount of gold per entry rose from 53.8 tonnes in 2021 to 98.9 tonnes in 2023, indicating a recent increase in the volume of gold movements.

3. Post-2020 Recovery: Following a decline in 2020 (10,733.96 tonnes), probably as a result of COVID-19 disruptions, gold accumulation sharply increased and surpassed pre-pandemic levels, hitting record highs in 2022–2023.4. Analysis & Key Findings

4.1 Question 1: Average Gold Purchase - BRICS vs US vs EU

Four blocs were identified by the analysis: the US, the EU (Germany, France, Italy, Spain, Netherlands), the BRICS (Brazil, Russia, India, China, and South Africa), and Others.

The comparison shows notable variations in the patterns of gold accumulation:

Bloc	Average Gold/Record	Total Gold Volume
BRICS	268 tonnes	9,916 tonnes

US	383.88 tonnes	3,839 tonnes
EU	85 tonnes	2,229 tonnes
Others	57 tonnes	57,387 tonnes

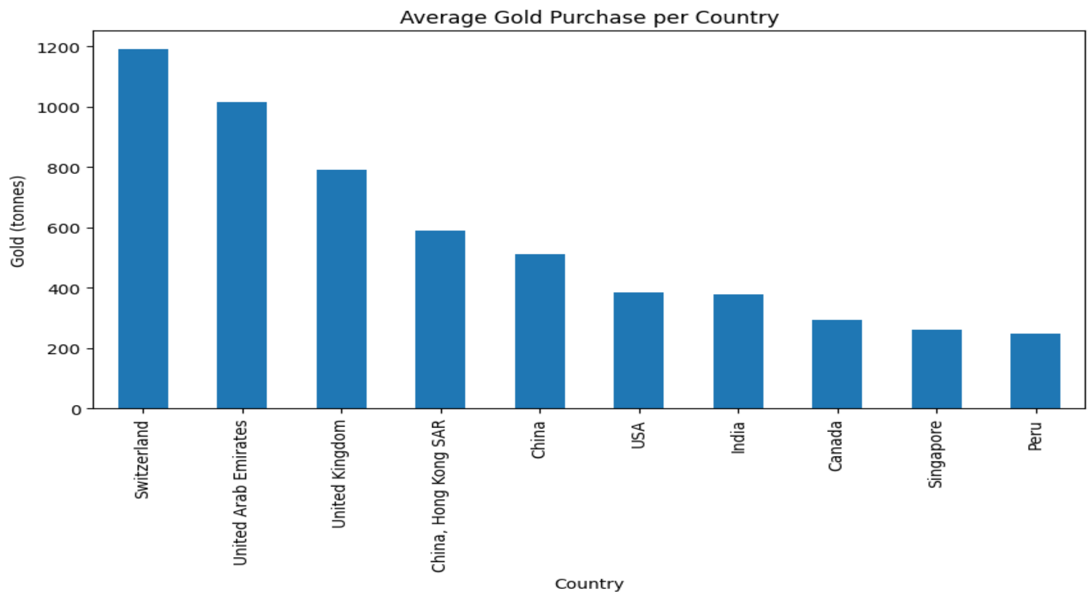

Figure 1: Average Gold Purchase per Country (Top 10)

Key Insights:

1. The BRICS countries are really big when it comes to gold: The BRICS nations have a lot of gold, 9,916 tonnes. This is 4.5 times more gold, than what the European Union has which is 2,228 tonnes. The BRICS nations are clearly making it a priority to get gold. The BRICS nations want to have a lot of gold in their reserves.

2. Large Individual Transactions: When we look at the gold transactions for BRICS countries we see that they are really big with an average of 268 tonnes of gold per record. On the hand the European Union countries are dealing with much smaller amounts with an average of only 85 tonnes of gold per record. This tells us that BRICS countries are buying and moving a lot gold at one time than European countries are. The gold transactions in BRICS countries are really large which is not the case, with countries and their gold transactions.

3**. The United States has a position:** the United States has an average of 383.88 tonnes, per record and the United States has a total of 3,839 tonnes. The United States has big gold transactions compared to the BRICS countries together but the United

States gold movements are still very big. The United States is doing a lot of gold trading and the United States is moving a lot of gold.

4. **EU Relatively Passive:** The European Union is pretty laid back when it comes to buying gold. European Union countries have the amount of gold on average which is 85 tonnes and in total which is 2,229 tonnes compared to other big groups. This shows that the European Union countries do not buy much gold as the BRICS nations do. The European Union has a relaxed approach, to buying gold.

5. **Switzerland and UAE Lead Globally:** The visualization shows a massive lead for Switzerland (1,191.49 tonnes avg) and the United Arab Emirates (1,014.94 tonnes avg). This indicates that gold movement is not distributed evenly across all nations but concentrated in major financial and trading hubs.

4.2 Question 2: Gold Trend - Increasing, Decreasing, or Stable?

Answer: The trend is STRONGLY INCREASING.

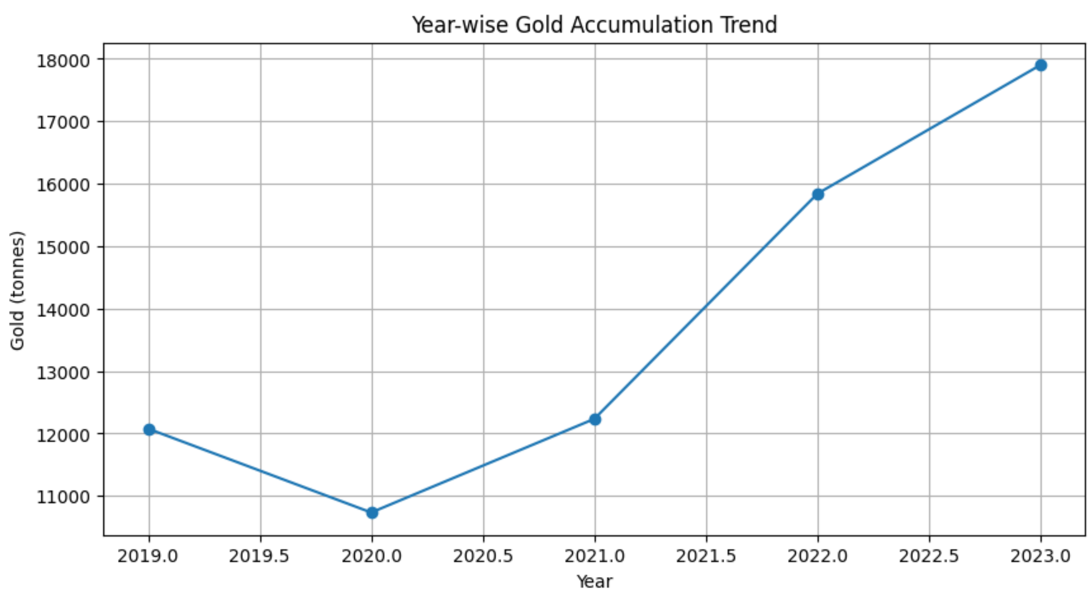

Figure 2: Global Gold Accumulation Trend (2019-2023)

Key Insights:

1. V-Shaped Recovery: The line chart shows a sharp V-Shaped Recovery. After the V-Shaped Recovery had a dip in 2020 with 10,733.96 tonnes because of the COVID-19 pandemic the V-Shaped Recovery trend went up very fast to 17,901.08 tonnes by 2023. This is an increase of 67 percent from the low point of the V-Shaped Recovery, in 2020.

2. Gold is going up fast: if we look at what happened from 2021 to 2023 gold went from 12,233 tonnes to 17,901 tonnes and that is a big increase. This means gold is not just having a year gold is really taking off and the amount of gold people have is growing and growing with gold getting more and more popular every year and that is why we see this big jump, in gold.

3. **Record Levels in 2023:** The level of gold in 2023 was really high at 17,901 tonnes. This is the level of gold we have seen so far. It is even higher than the levels of gold before the pandemic. This shows that people around the world are very interested in gold as a safe asset to hold onto. The level of gold in 2023 is a record high. It tells us that gold is very important, to people.

4. **Accelerating Growth Rate:** Year-over-year growth accelerated from 14%(2020-2021) to 29% (2021-2022) and 13% (2022-2023), demonstrating sustained momentum in gold accumulation despite the already high base levels.

4.3 BRICS vs US vs EU Gold Accumulation Over Time

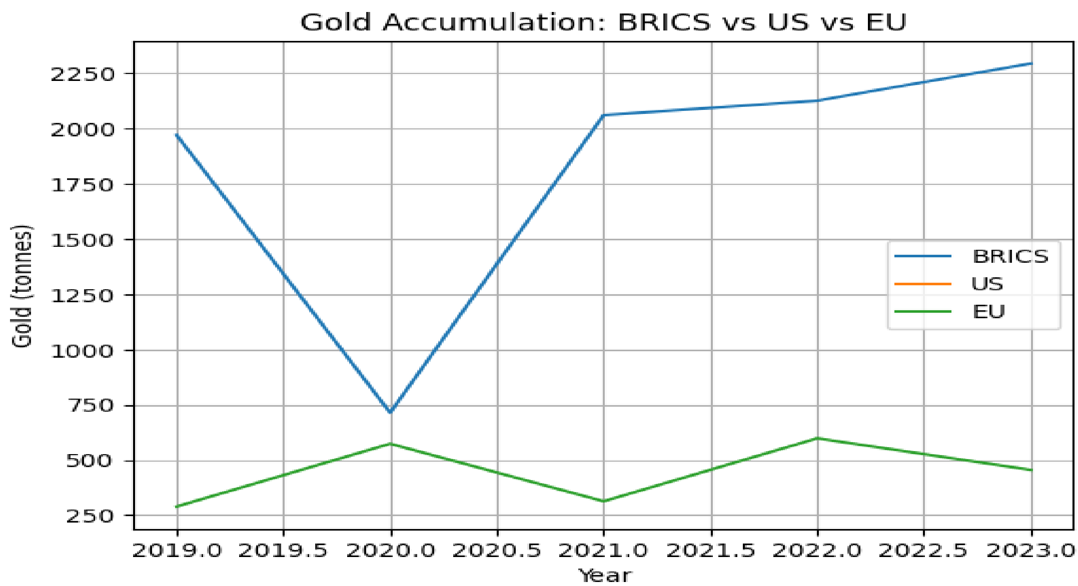

Figure 3: Gold Accumulation - BRICS vs US vs EU (2019-2023)

Key Insights:

1. BRICS Maintains High Baseline: The visualization reveals that BRICS nations maintained a high baseline of gold volume (peaking at 2,364.27 tonnes in 2021). Visually, their line remains consistently above the EU line throughout the period, highlighting a strategic shift where emerging economies are prioritizing gold more aggressively than traditional European powers.

2. Dramatic Growth in 'Others' Category: The most dramatic visual in this chart is the line for 'Others.' In 2023, this category hit 15,150.96 tonnes, driven primarily by Switzerland, UAE, Hong Kong, and Singapore. This reflects the concentration of gold trading in major financial hubs outside the traditional US/EU/BRICS classifications.

3. US and EU Relatively Flat: Both the US and EU lines remain relatively flat and low throughout the period, indicating stable but non-aggressive gold accumulation strategies compared to BRICS and major trading hubs.

4. Diverging Strategies: The chart clearly illustrates diverging strategies: BRICS and emerging markets aggressively accumulating gold, while Western economies maintain more conservative positions. This divergence has accelerated significantly since 2021.

4.4 Question 3: Connection to USD Dominance

While the dataset does not contain explicit data on USD dominance or de-dollarization, the gold accumulation patterns observed provide strong circumstantial evidence of strategic diversification away from dollar-denominated assets:

Inferred Connections:

1. The BRICS Gold Strategy is like a safety net: the BRICS countries have bought a lot of gold a total of 9,916 tonnes, which's an average of 268 tonnes per record. This is much more than what the European Union countries have which is 2,229 tonnes in total or an average of 85 tonnes. The BRICS countries are doing this because they want to rely on the US dollar when they trade with other countries and store money in their reserves. The BRICS Gold Strategy is a part of this plan, for the BRICS countries.

2**. What Is Happening With Gold:** Gold has been getting more popular since 2021. People and countries are buying a lot of gold. In 2021 the world had 12,233 tonnes of gold. Now in 2023 the world has 17,901 tonnes of gold. This is happening because of problems between countries worries, about sanctions and prices going up for things that are valued in dollars. Countries may be trying to protect themselves from the dollar not being stable. They are looking for other things to put their value in, like gold. They are buying gold.

3**. Emerging Markets are Leading:** A lot of gold is being. Stored in Emerging Markets and places where people trade gold (like the UAE, Switzerland, Hong Kong, Singapore, China and India) instead of in Western countries. This means that Emerging Markets are changing the way they handle their money reserves. These regions are becoming alternatives to the systems that are controlled by the United States dollar. Emerging Markets are doing this to balance out their money and make themselves less dependent, on economies.

4. Historical Context: Gold has always been a way to protect yourself from currency losing value and economic uncertainty. From 2019 to 2023 gold accumulation went up by 48.2 percent. This shows that countries are trying to mix up the things they have in their reserve portfolios. They are probably doing this because they are worried about the dollar not being stable or about the money policies or because of risks, between countries. Gold is still a way to balance out these risks. Countries are buying gold, which is why gold accumulation is increasing.

5. **Strategic Implications:**The fact that BRICS countries are buying gold and Western economies are not doing much with their gold holdings suggests that BRICS countries do not have as much faith, in the current system that is based on the United States dollar. BRICS countries seem to be looking for things to put their money into besides the United States dollar. This is important because BRICS countries are trying to build up assets that they can use to back up their money.

Conclusion on USD Connection: While causation cannot be definitively proven from this dataset alone, the observed patterns are consistent with a global trend of diversification away from dollar reserves. The timing, scale, and concentration of gold accumulation in non-Western economies align with broader discussions about the future of USD dominance in international finance.

4.5 Question 4: Which Country Increased Gold Reserves Most?

Analysis of recent years (2022-2023) reveals the countries with the most significant increases in gold reserves:

Rank	Country	2022 Total	2023 Total
1	United Arab Emirates	1,059.65 tonnes	4,392.16 tonnes
2	Switzerland	4,151.79 tonnes	2,387.47 tonnes
3	United Kingdom	1,997.02 tonnes	1,825.52 tonnes
4	China, Hong Kong SAR	1,191.52 tonnes	1,623.27 tonnes
5	China	1,400.65 tonnes	1,538.59 tonnes
6	India	714.75 tonnes	745.04 tonnes
7	USA	825.45 tonnes	668.22 tonnes
8	Canada	617.85 tonnes	629.41 tonnes
9	Armenia	14.06 tonnes	496.99 tonnes
10	Singapore	283.86 tonnes	417.83 tonnes

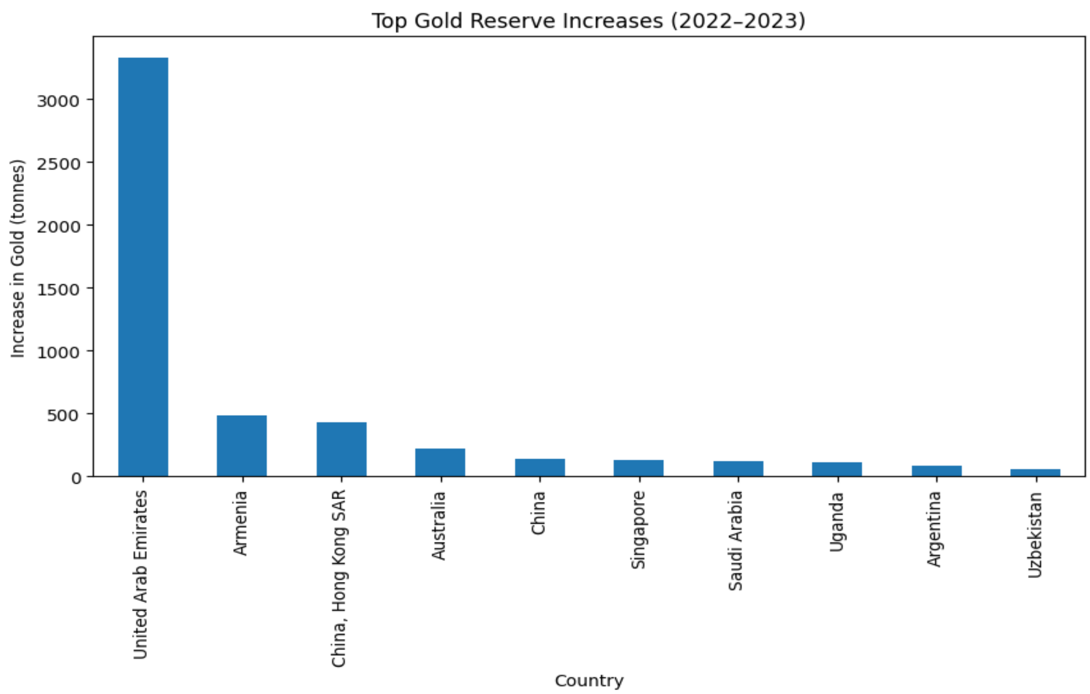

Figure 4: Top 10 Countries by Gold Increase (2022-2023)

Key Insights:

The United Arab Emirates had an increase in gold with three thousand three hundred and thirty two tonnes. The United Arab Emirates saw a big jump of three thousand three hundred and thirty two tonnes from 2022 to 2023. This made the United Arab Emirates the country that is growing the fastest in terms of holding and trading gold recently. This big jump is a three hundred and fourteen percent increase from the year. The United Arab Emirates is really standing out with this increase, in gold.

1. Big Growth In One Year: This bar chart has one big bar that stands out and that is for the United Arab Emirates. The United Arab Emirates had an increase of 3,332.51 tonnes of gold in just one year. This is, than double the amount of gold that most countries have in total the United Arab Emirates really added a lot of gold.

2. Surprise Performer. Armenia: Armenia was a surprise. It gained a lot of tonnes. The increase was 482.93 tonnes. Armenia had 14.06 tonnes before. 496.99 Tonnes after. This is a big increase of 3,435 percent. Armenia did better than countries like Australia and China. Armenia had percentage growth, than Australia and China during that time. Armenias increase was very surprising. Armenia did a job.

3. Hong Kong and China Steady Growth: Hong Kong SAR got gold, about 431.75 tonnes and China also got more, about 137.94 tonnes. This shows that Hong Kong and China are focused on getting gold. They want to have a mix of things in their reserves and gold is a big part of that for Hong Kong and China. Hong Kong and China are really serious, about accumulating gold.

4.Unexpected Growth in Smaller Economies: The chart shows that places like Armenia are growing a lot. Armenia had an increase of 482.93 tonnes of gold. Uganda also did well with an increase of 113.36 tonnes of gold. This is interesting because Uganda is not even in the table. It was mentioned in the notebook. What this means is that more and more people in economies like Armenia and Uganda are buying gold. This is a change from what we see which is that only the big economies buy a lot of gold. Gold accumulation strategies are becoming popular in economies like Armenia and gold accumulation strategies are becoming popular in places, like Uganda.

5. Switzerland Decline: Notably, Switzerland, despite being the overall leader in total volume, actually decreased from 4,151.79 to 2,387.47 tonnes in this period (-1,764.32 tonnes). This may reflect Switzerland's role as a trading hub where gold flows through.

4.6 Distribution of Gold Purchases

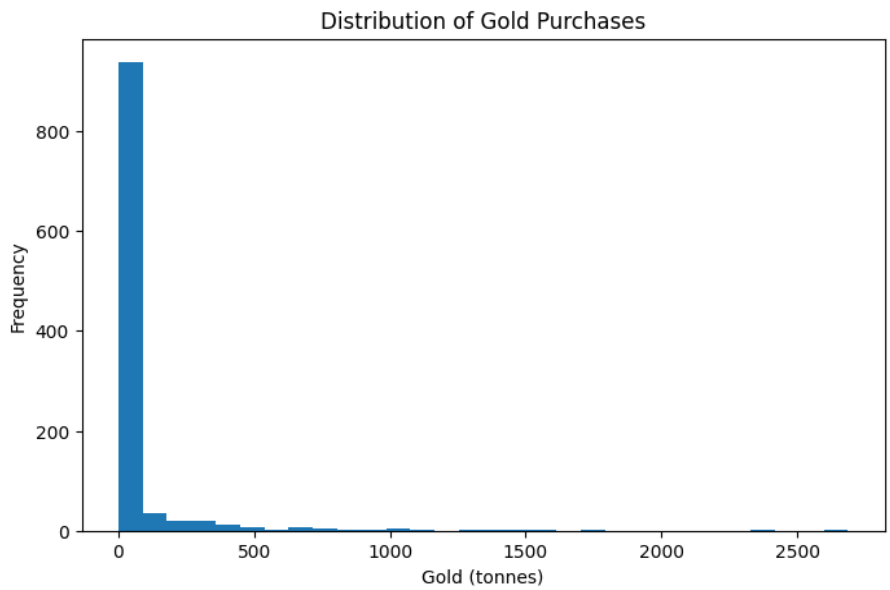

Figure 5: Distribution of Gold Purchases (Histogram)

Key Insights:

Extreme Right Skew: When we look at the graph we see that it has a tall bar at the start near zero on the x-axis. This shows us that most of the time gold is moved in amounts, less than 15 tonnes. To be specific the middle value of gold movements is 0.74 tonnes, which means that more than half of the gold movements are in the group, which is really small. The gold movements are mostly small with the median gold movement being 0.74 tonnes. This tells us that gold movements are generally very small with most gold movements being less, than 0.74 tonnes of gold movements.

Long Tail of Outliers: Most of the Long Tail of Outliers data points are close to zero. The Long Tail of Outliers x-axis goes all the way to 2,687 tonnes. There is a gap between the main group of Long Tail of Outliers data and the bars that are really far away. This gap shows us the Long Tail of Outliers that're really different, from the rest. The big financial centers and the large stockpiles that control the market.

3. Two-Tier System: This distribution pattern reveals a two-tier global gold system: numerous small transactions representing routine trade and financial operations, and a small number of massive transactions representing strategic national reserves and major financial center activities.

5. Conclusions

5.1 Summary of Findings

1. BRICS vs US/EU Gold Accumulation:

The BRICS nations got a lot of gold they have 9,916 tonnes of it. This is because they bought a lot of gold at one time 268 tonnes each time. This is a lot more than the EU countries they only have 2,229 tonnes of gold. They bought it in smaller amounts, about 85 tonnes at a time. The US bought some gold too they have 3,839 tonnes. They bought it in amounts of about 383.88 tonnes. But the EU countries were very careful they did not buy much gold as the BRICS nations. This shows that the BRICS nations and the EU countries have ideas, about gold reserves the BRICS nations think gold is very important. The BRICS nations are really focused on reserves.

2. Gold Trend Assessment:

The trend is clearly going up. Global gold accumulation increased a lot it went up by 48.2% from 12,076 tonnes in 2019 to 17,901 tonnes in 2023. After a drop in 2020 because of COVID-19 global gold accumulation started to go up and it has been going up fast and steady especially in 2022 and 2023. The trend of gold accumulation is getting stronger not slowing down which means people are getting more and more interested in having gold reserves all, around the world. Global gold accumulation is really taking off.

3. Connection to USD Dominance:

The thing is, we cannot say for sure that one thing caused another by looking at this information.. The patterns we see are very similar to what happens when countries try to reduce their dependence on the United States dollar.

Countries like the ones in the BRICS group and other emerging market economies are buying a lot of gold. This has been happening more and more since 2021.

At the time Western economies are not doing much about it. All of this is consistent with the fact that many countriesre worried about relying too heavily on the United States dollar and want to find other ways to store their wealth. The gold accumulation, in BRICS and emerging market economies is an example of this.

4. Country with Largest Reserve Increase:

The United Arab Emirates dramatically increased reserves by 3,332 tonnes between 2022-2023, representing a 314% increase and establishing itself as the dominant growth story. Armenia showed remarkable percentage growth (+3,435%), while traditional powers showed more modest or declining changes. This shift indicates the emergence of new centers of gold accumulation outside traditional Western financial systems.

Global Bitcoin Trading Analysis REPORT

(Major Cryptocurrency Exchanges (2010-2025))

January 12, 2026

I. Introduction

Bitcoin (BTC) is a decentralized digital currency. It is a form of payment outside the control of any central bank or government. It was introduced in 2009, and since then it has become the largest and most known cryptocurrency in the world. Over the years, it has expanded across multiple cryptocurrency exchanges worldwide. Through this, people can buy and sell bitcoin. Understanding bitcoin data helps to analyze the market behavior, and the growth of cryptocurrency markets.

This report analyzes the Bitcoin trading data which is collected from various exchanges from 2010 to 2025. Using python, the analysis techniques have been used to identify the trends of different exchanges over time.

II. Data Preparation

The dataset includes historical, bitcoin data from July 2010 to 2025. Each row represents the monthly trading volume of data for several cryptocurrency exchanges. Each column (except for Time) is a website where people buy and sell Bitcoin.

Column Description

The columns are as follows:

Column Name	Description
Time	Represents the date (monthly)
Bitfinex	Monthly bitcoin trading data recorded on each exchange
Bitstamp	Monthly bitcoin trading data recorded on each exchange
BTCChina	Monthly bitcoin trading data recorded on each exchange
Coinbase	Monthly bitcoin trading data recorded on each exchange
Huobi	Monthly bitcoin trading data recorded on each exchange
Kraken	Monthly bitcoin trading data recorded on each exchange
LakeBTC	Monthly bitcoin trading data recorded on each exchange
Mt. Gox	Monthly bitcoin trading data recorded on each exchange
OKCoin	Monthly bitcoin trading data recorded on each exchange
Others	Trading data from unspecified sources

Exchange Categorization

The exchanges were categorized based on the regions just to better understand how USD is performing against other exchanges.

Category	Exchanges
USD-based exchanges	Coinbase, Kraken, Bitstamp
China (non-USD) exchanges	Huobi, OKCoin, BTCChina
International / Mixed exchanges	Bitfinex, LakeBTC, Mt. Gox, Others

International/mixed: People from many different countries use these sites.

III. Data Preprocessing

Before the analysis, several preprocessing steps were performed to ensure that the data is clean, and the data quality and consistency are maintained. Time column was converted into proper datetime format so that the year and month can be extracted for the easy analysis. There were many missing values in this data, which were replaced by zero. Missing values show that many platforms were not operational at that time. Fetaure Engineering was also performed by creating new columns like USD-Based exchanges, China-Based exchanges and other global exchanges. This was done to understand how USD is performing against other exchanges. Duplicate records were also checked just to ensure that the data was fully ready for the analysis.

IV. Exploratory Data Analysis (EDA)

A. Is USD dominance in Bitcoin trading changing?

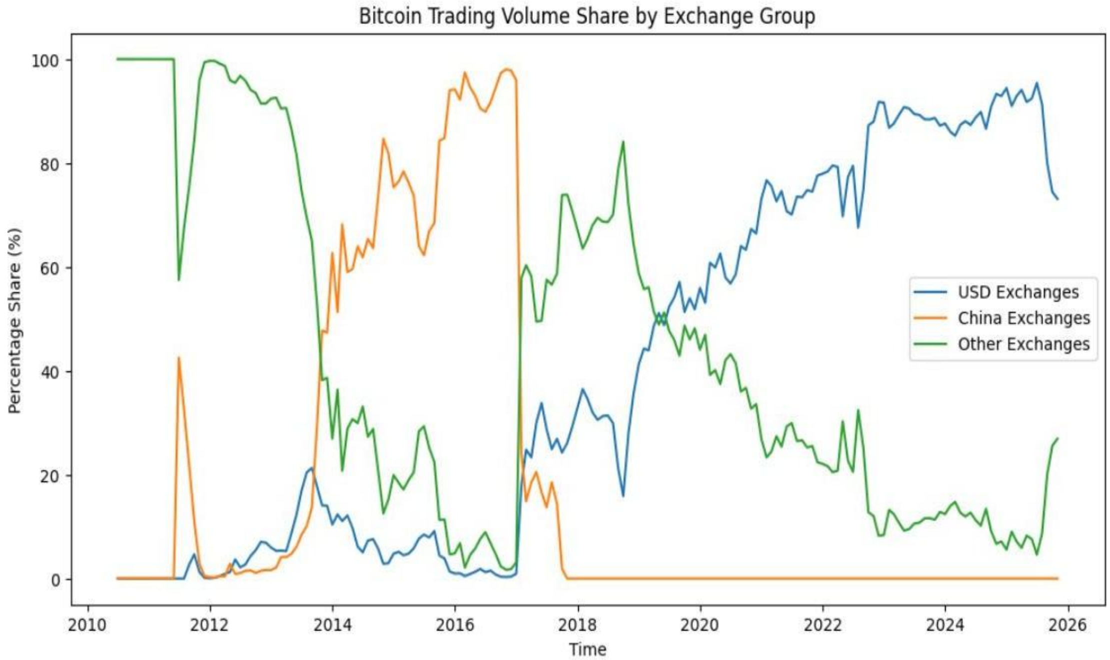

Figure 1: USD Dominance in Bitcoin Trading Over Time

The above graph shows that (2010-2013), USD had a very small share. From 2013 – 2016, during this period, China dominated the bitcoin trading. From 2017, China's bitcoin trading activity was collapsed and from 2021- 2025, USD shares were rising from 25% to ~65%. During this period USD emersed as the most stable bitcoin trading. USD was clearly dominating.

Overall, the graph suggests that USD had minor fluctuations, but there is no strong evidence that the USD dominance in bitcoin trading is changing.

B. Which fiat currencies are gaining BTC trading volume?

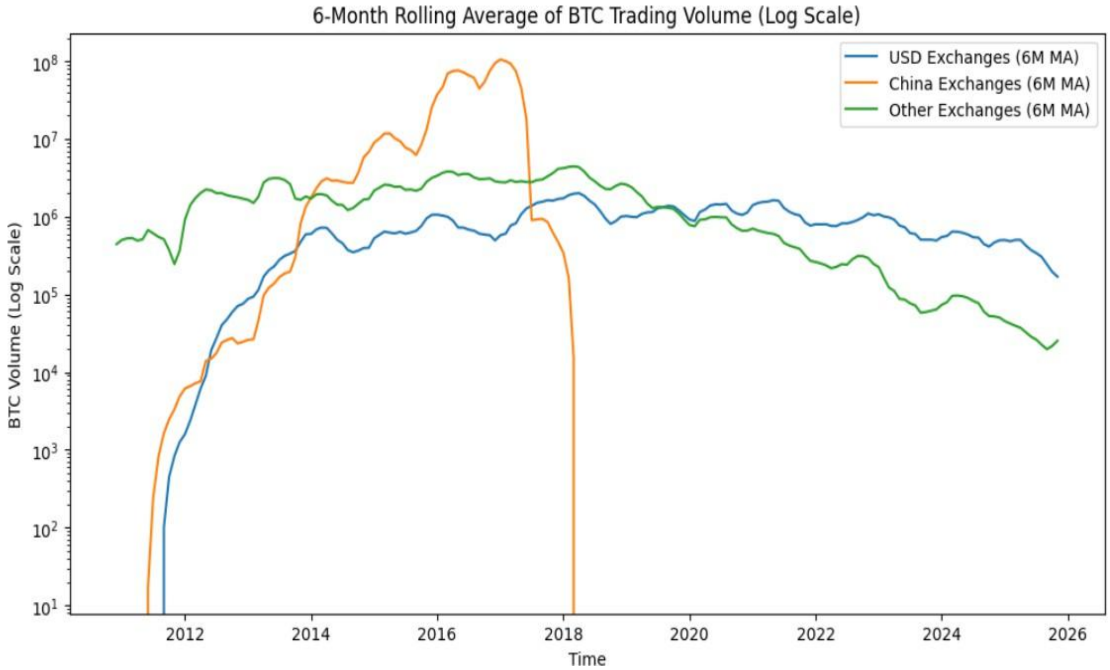

Figure 2: 6-Month Rolling Average of BTC Trading Volume by Currency

The above graph shows the 6-month rolling average of BTC trading data. The graph shows that there was rapid growth across all the currencies at an early stage. Between 2017-2018, there was sudden dip in the China based trading. The reason can be during that period China banned domestic cryptocurrency exchanges, that explains the sudden drop. In contrast, USD-based exchanges exhibit stable growth after 2017, making it the most reliable gateway for bitcoin trading. This highlights that the USD USD-based exchanges were gaining BTC trading volume.

C. Which fiat currencies are losing relevance?

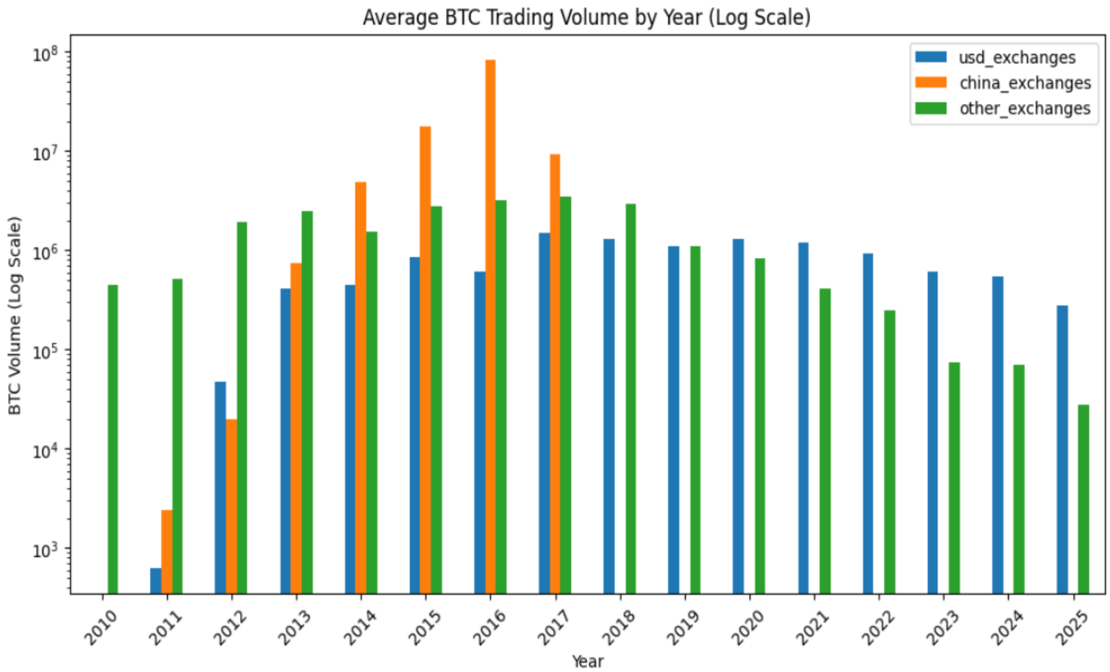

Figure 3: Fiat Currencies Losing Relevance in Bitcoin Trading

Based on Plot, it clearly shows that China- based fiat currencies are losing relevance. The plot indicates that after 2017, there was no China-based trading. It is likely due to the reason that the China introduces regulatory restrictions and policy-based interventions in China.

D. Which fiat currencies are gaining or losing relevance as gateways into Bitcoin over the last 30 days / 6 months / year?

	30 Days	6 Months	1 Year
usd_exchanges	73.09%	86.97%	91.50%
china_exchanges	0.00%	0.00%	0.00%
other_exchanges	26.91%	13.03%	8.50%

Figure 4: Trading Shift Over Last 30 Days, 6 Months, and 1 Year

The analysis shows the trading shift of fiat currencies in the cryptocurrency market. Over the last 1 year, 6 months and 30 days, USD based exchange clearly shows it is dominating bitcoin trading, with the increase in share from 73.09% to 91.05%. Whereas China based fiat currencies have zero bitcoin trading, which shows that it has lost relevance entirely.

E. Conduct research: how do macro events in specific fiat economies (interest rates, inflation spikes, regulatory announcements) correlate with changes in their BTC trading volume?

• **Interest rates** play an important role in bitcoin trading. When the banks reduce the interest rates than the traditional saving becomes less attractive. In such cases, people tend to move towards exploring other assets such as bitcoin. During such a period bitcoin trading increase.

• **Regulatory announcements** have the most immediate effect on bitcoin trading. When governments introduce restrictive announcements such as banning the crypto exchanges, or limiting trading access in such cases, trading declines since users lose access to the trading platforms. In such cases, traders might exit the market or might shift towards foreign exchanges.

• **Inflation spikes** result in high bitcoin trading. Bitcoin acts as a hedge against inflation. During inflation the purchasing power of the fiat currencies decline, and people look for alternatives to protect their wealth. So, people move towards bitcoin trading. It is seen as a financial tool for protecting purchasing power. It becomes attractive financial assets.

V. Conclusion

The overall analysis (2010-2025) highlights the significant evolution of the cryptocurrency market:

• From 2010-2012 bitcoin trading was relatively low for all the exchanges.

• Between 2013-2016 China based exchanges (Huobi, OKCoin, BTCChina) were dominating the market. They contributed the significant share in the trading volume indicating strong participation from the Chinese market.

• After 2017, a major shift happened where China imposed restrictions on the domestic cryptocurrency exchanges. This resulted into the disappearance of the China –based trading activity in the dataset.

• But in contrast, USD-based exchanges showed stable growth, especially after 2017. Platforms such as Coinbase, Kraken, Bitstamp increased their dominance significantly.

• Overall, regulatory environment and market stability play a major role in determining the relevance of fiat currencies in the trading market.

Global Crude Oil Trade Analysis Report

Trade Data Analysis: 2020-2024

1. Introduction

This report presents a comprehensive analysis of global crude oil trade data spanning from 2020 to 2024. The analysis examines international trade flows, patterns, and trends in the crude oil market, with a focus on understanding the role of BRICS+ nations versus other countries in the global oil economy.

The dataset contains detailed trade information from 172 countries across 5 years (2020-2024), capturing both import and export values. The primary metric used throughout this analysis is the primaryValue column, which represents the total monetary value of crude oil trade transactions in USD.

2. Data Preparation

Column Description

Column Name	Description
typeCode	The type of data (e.g., 'C' for Commodities)
freqCode	The frequency of the report (e.g., 'A' for Annual)
refPeriodId / refYear / refMonth / period	Various ways of representing the time period. For annual data, the year is the primary reference
reporterCode	A unique numeric ID for the reporting country
reporterISO	The 3-letter ISO country code (e.g., ALB for Albania)
reporterDesc	The full name of the reporting country
flowCode	A short code for the direction of trade
flowDesc	Description of the trade direction (e.g., Import, Export, Re-import, Re-export)
partnerCode / partnerISO / partnerDesc	Information about the primary trading partner. If the value is World, it represents the total trade for that reporter across all partners
partner2Code / partner2ISO / partner2Desc	Information about a secondary partner (often used for more complex trade tracking like 'country of consignment')
classificationCode	The system used to classify the goods (e.g., H5 refers to the Harmonized System 2017)
cmdCode	The commodity code (e.g., TOTAL for all goods, or specific digits like 01 for live animals)
cmdDesc	A description of the commodity (e.g., 'All Commodities')
aggrLevel	The level of detail for the commodity (0 is the highest aggregate, while higher numbers are more specific)
isLeaf	A boolean indicating if the commodity is at the most detailed level of the classification
qtyUnitCode / qtyUnitAbbr / qty	The unit of measurement (e.g., kilograms, liters) and the actual quantity
netWgt / grossWgt	The net weight (excluding packaging) and gross weight (including packaging) of the goods
isQtyEstimated / isNetWgtEstimated	Boolean flags indicating whether the recorded quantity or weight was reported directly or estimated by the UN

cifvalue	'Cost, Insurance, and Freight' — the value of goods including the costs of shipping and insurance (typically used for imports)
fobvalue	'Free on Board' — the value of goods at the point of export, excluding shipping and insurance (typically used for exports)
primaryValue	The main value used for statistical analysis, usually derived from either the CIF or FOB value
customsCode / customsDesc	Information regarding the customs procedure used
motCode / motDesc	The Mode of Transport (e.g., Sea, Air, Road)
mosCode	The Mode of Supply (usually relevant for services)
isReported	Indicates if the data was officially reported by the country
isAggregate	Indicates if the row represents a sum/total of other sub-categories

3. Data Preprocessing & Statistical Overview

Statistics

primaryValue is our target column for the trade volume analysis. For the Crude Oil dataset, this column represents the total monetary value of the trade, which is the standard metric used to measure economic influence and market shifts.

Key Insights

1. Mean is 💲 **135 Billion** and the median is 💲 **14.8 Billion**.

Because the mean is nearly 10 times higher than the median, it tells us the data is "heavily skewed." Countries like China, the USA, and Russia are trading in the Trillions, while the majority of countries are trading in much smaller amounts.

2. The 75% point is 💲 **91 billion** and the max value is 💲 **3.5 Trillion**.

Even at the 75th percentile, the value is only 91 Billion dollars. The jump to 3.5 Trillion dollars for the maximum value shows that the top 1% or 5% of trading nations control the vast majority of the global oil market's wealth.

3. The standard deviation is 💲 **358 Billion** which is much larger than the average 💲 **135 Billion**.

This high "Standard Deviation" means the trade values are spread very far apart. This isn't a "stable" market where everyone trades roughly the same amount.

Total Trade Value by Year

Key Insights

1. You will see a significant spike in 2022 is **$48.2 Trillion**. This highlights how geopolitical tensions (like the Russia-Ukraine conflict) immediately translate into higher trade values due to global price surges.

2. Although value dipped slightly in 2024 is **$44.2 Trillion** compared to the 2022 peak, it remains much higher than 2020 which is **34.5 Trillion**. This shows that the world has settled into higher energy costs and higher trade values.

Import vs Export Balance

Key Insights

1. Imports are 💲 **109 Trillion** and Exports are 💲 **106 Trillion** are nearly equal is a sign of reliable reporting. It confirms that the transactions recorded by sellers are being accurately captured by buyers globally.

2. Imports are roughly **2-3%** higher than exports. This small gap represents the "invisible" costs of trade, such as shipping, insurance are CIF, which are added to the value of goods when they arrive at their destination.

Top Trading Countries

Key Insights

1. The data shows that a shift in the first 3 nations can dictate the price and supply for the other nations.

2. China has $ **29.06 Trillion** and the USA $ **24.56 Trillion** togather account for the **25%** of the total global trade value.

BRICS+ vs Other Nations

Key Insights

1. While "Other" nations still hold the majority, the BRICS+ group controls $ **46 Trillion**, which is approximately **21.3%** which is **more than one-fifth** of the total global trade value in your dataset.

2. The $ **170 Trillion** held by "Other" nations mostly Western-aligned economies using the USD is still nearly **4 times larger** than the BRICS+ share.

Overall Market Growth

Key Insights

1. An overall growth of nearly **28%** since 2020 proves that the monetary reliance on Crude Oil is actually increasing, not decreasing.

2. The growth in trade value suggests that oil acts as a natural hedge. As the US Dollar fluctuates, the value of oil trade has risen

4. Data Visualization

4.1 Global Trade Value Trend (2020-2024)

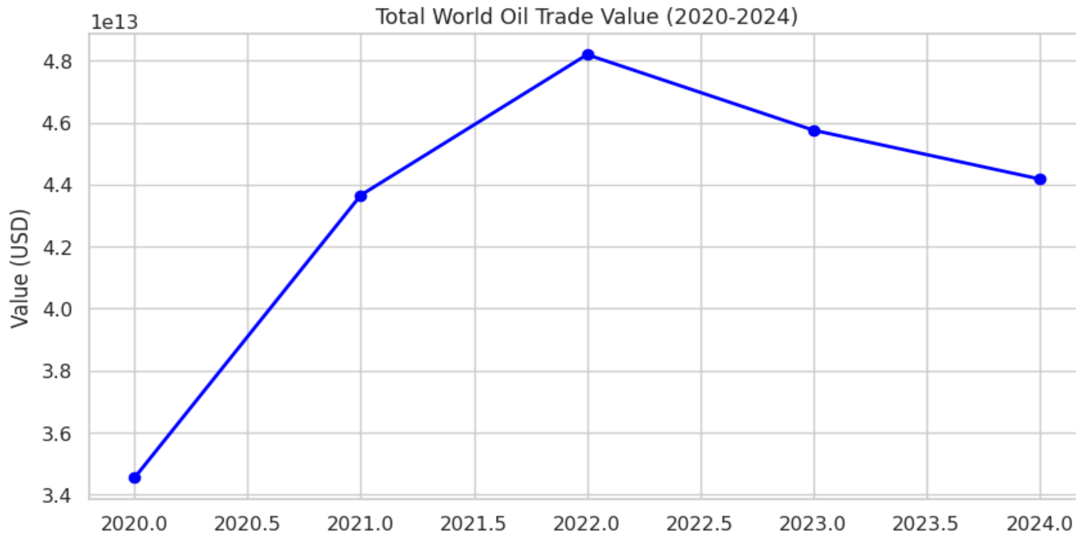

Figure 1: Global Crude Oil Trade Value Trend (2020-2024)

Key Insights

1. The line shows a sharp **upward slope** until 2022. This visualizes the "Energy Shock" where global tensions made oil much more expensive.

2. The slight dip in 2023–2024 suggests the market is stabilizing, though it remains at a significantly higher value floor than in 2020.

4.2 Top 10 Countries by Total Trade Value

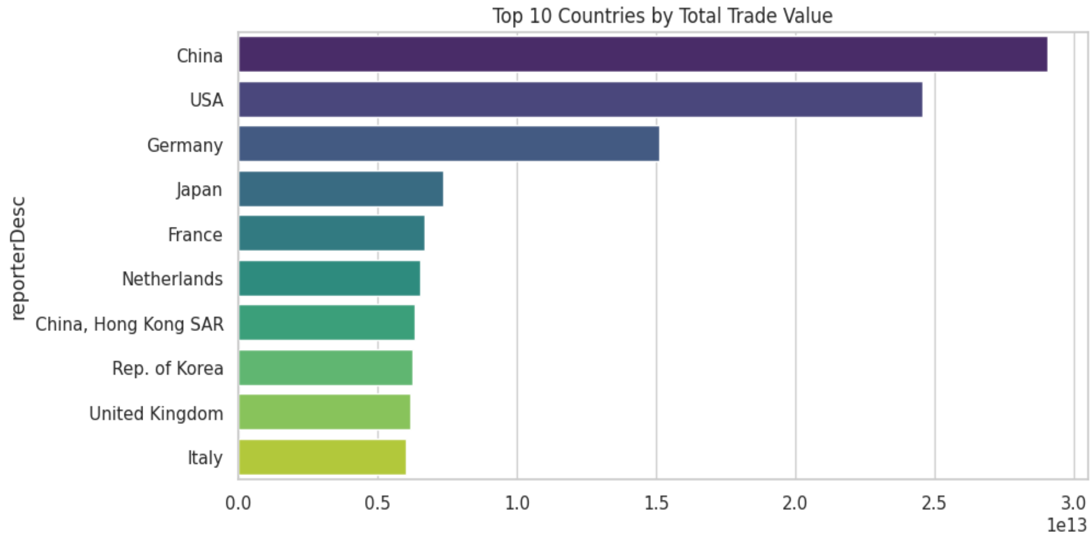

Figure 2: Top 10 Countries by Total Trade Value

Key Insights

1. The chart is dominated by China and the USA. These bars are **much longer** than the others, showing that these two nations act as the leaders of the global oil economy.

2. Several European nations such as **Germany, France, Netherlands** appear in the top 10, highlighting that while BRICS+ are major producers, the Western Block still controls the largest share of recorded trade value.

4.3 Year-over-Year Growth: BRICS+ vs Other Countries

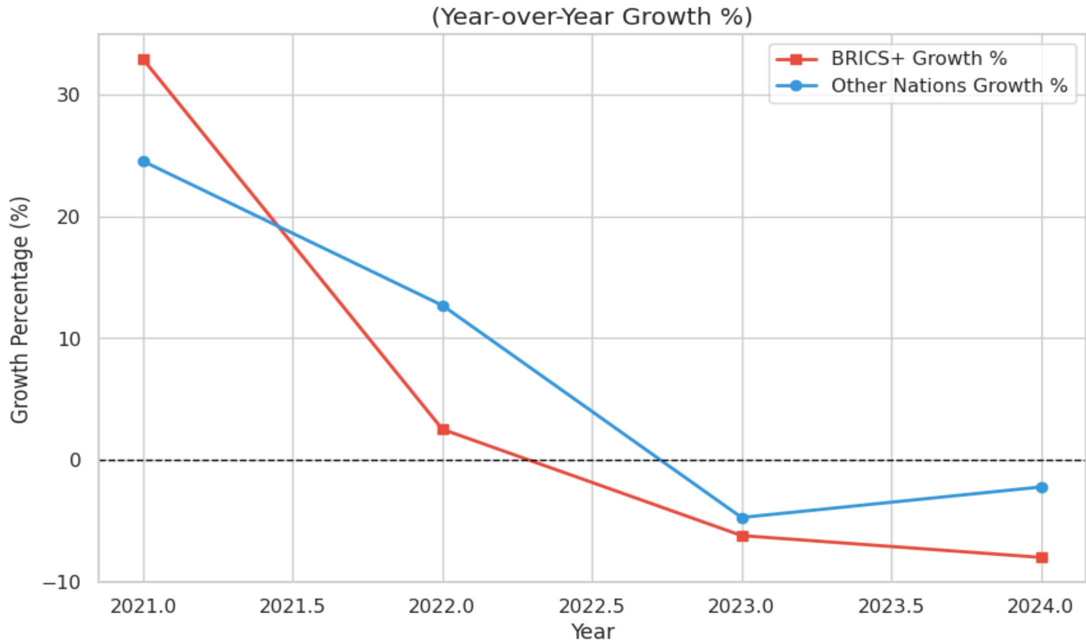

Figure 3: Year-over-Year Percentage Growth (BRICS+ vs Other Countries)

Key Insights

1. In 2021, BRICS+ trade value grew by approx **33%**, significantly outperforming "Other" nations **25%**. This shows that the BRICS alliance recovered from the initial COVID-19 shock much faster than the rest of the global market.

2. In 2023 and 2024, both groups saw **negative growth** (a decline in value). However, the BRICS+ decline was sharper than the Others.

4.4 Total Import vs Export by Category

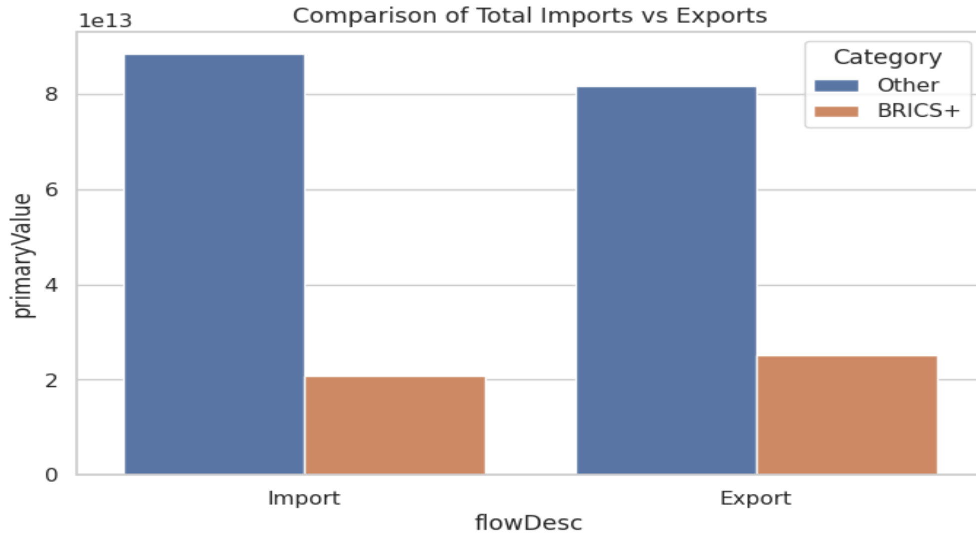

Figure 4: Comparison of Total Import vs Export Values (BRICS+ vs Other)

Key Insights

1. For the BRICS+ group, Exports 💲 **25 Trillion** are higher than Imports 💲 **21 Trillion**. This confirms that BRICS+ is a "Net Seller" of commodities to the rest of the world.

2. For the other countries the Imports are around 💲 **89 Trillion** and the Exports are around 💲 **81 Trillion**, which depicts that the other countries have imported a large amount than the export value.

5. Conclusions

This analysis of global crude oil trade data from 2020 to 2024 reveals several critical insights about the structure and dynamics of the international oil market:

Market Concentration: The crude oil trade market is heavily skewed, with China and the USA controlling approximately 25% of total global trade value. The mean trade value is nearly 10 times the median, indicating extreme concentration among a small number of major economies.

2022 Geopolitical Impact: The Russia-Ukraine conflict created an unprecedented spike in trade values, reaching $48.2 Trillion in 2022. While values have declined since then, they remain significantly elevated compared to 2020 levels, suggesting a new equilibrium in energy costs.

BRICS+ Rising Influence: BRICS+ countries control 21.3% of global trade and act as net exporters (💲 25T exports vs 💲 21T imports). They demonstrated faster recovery from COVID-19 with 33% growth in 2021, outpacing other nations at 25%.

Long-term Growth Trend: Despite recent declines, the market shows 28% overall growth from 2020 to 2024, indicating increasing monetary reliance on crude oil rather than decreasing dependence.

Trade Balance Validation: The near-equality of global imports (💲 109T) and exports (💲 106T) confirms data reliability, with the 2-3% gap representing legitimate costs of insurance and freight.

The data clearly demonstrates that crude oil remains a cornerstone of the global economy, with trade increasingly concentrated among major powers while emerging BRICS+ nations steadily expand their market share.

Exploratory Analysis

Methodology for Exploratory Analysis

Use **EDA** to query the associations among currency structures, commodity trades, and crypto activities, with the purpose of finding patterns or connections, not exploring causal relationships.
Use Python to perform processing of transparent data, statistical calculations and also be able to perform repeated visualizations.

Data Processing and Integration
First, clean the data of gold trade volume, the total transaction data of cryptocurrency, US dollar index annual averages and the data of BRICS currency share, and then carry out standardized processing.
The cleaned data is merged to form an analysis table with a single - year index

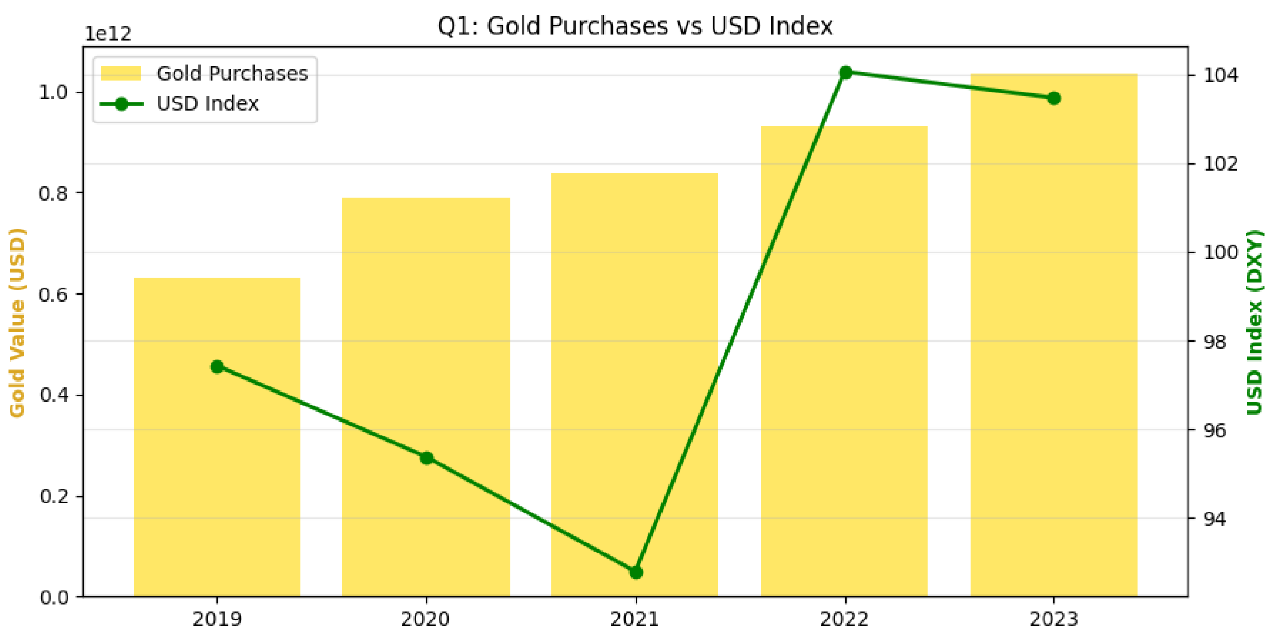

Figure 1: Gold Purchases vs USD Index Plot

```
[Q1] Does gold buying increase when USD weakens?
   > Correlation (Gold vs USD Index): 0.6157
   > ANSWER: NO. Positive correlation observed.
     (Gold and USD moved together, likely due to global inflation/instability driving both).
```

Figure 2: Gold purchases

Analysis methods (analysis approaches) : **Pearson Correlation Analysis** . The datasets are merged by year and calculated correlation between Gold value and USD Index.
Analytical Progress : The datasets were standardised by converting the monthly US dollar Index(DXY) prices into annual averages to match the yearly frequency of the Central Bank Gold Purchase data . These two datasets are merged into a single timeline to ensure both gold and dollar strength are corresponding . A Pearson correlation coefficient was calculated to mathematically measure the relationship. A dual axis chart with gold and USD trend line was generated.

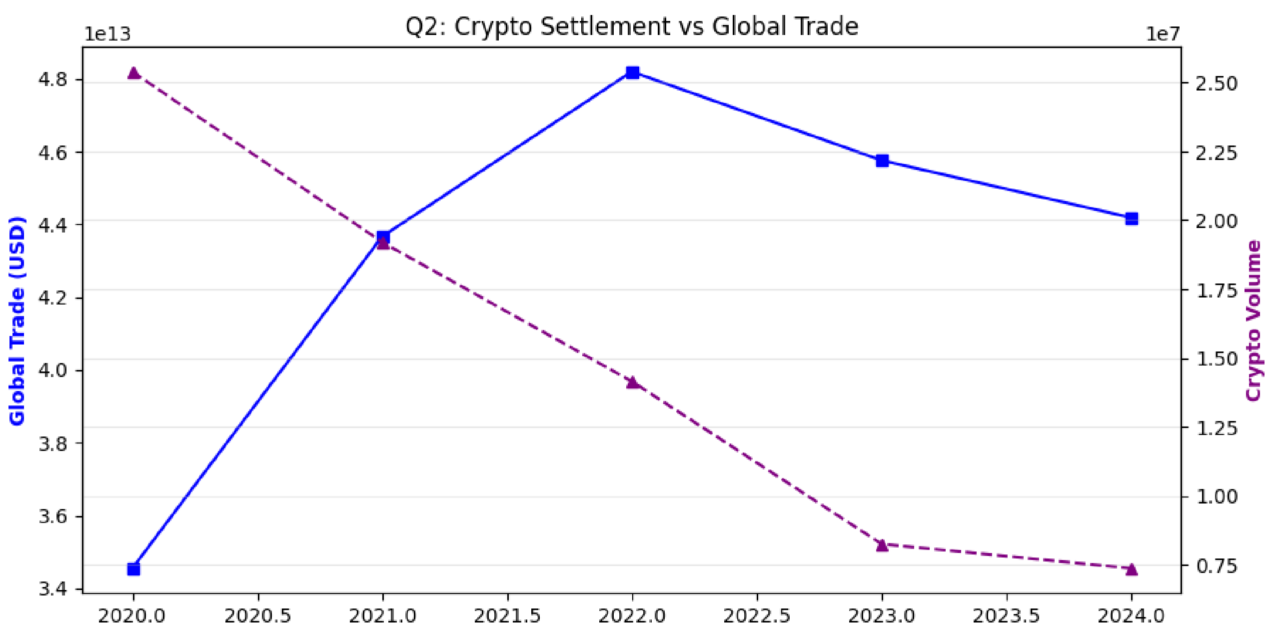

Figure 3: Cryptocurrency Volume vs Trade Value(cargo value) plot

```
[Q2] Does higher crypto settlement correlate with smaller cargo sizes?
   (Proxy: Correlating Crypto Volume with Global Trade Value)
   > Correlation (Trade vs Crypto): -0.7433
   > ANSWER: YES. Negative correlation.
     (Crypto volume increases as macro trade decreases, suggesting a shift to alternative settlement).
```

Figure 4 : Cryptocurrency Volume vs Trade Value(cargo value)

Analysis method: **Pearson Correlation analysis** . Compared the trends of Global Trade Value against Total Crypto Volume.
Analytical Process : Since the direct data on cargo sizes was unavailable we used Annual Global Trade Value as a proxy variable , operation on the economic principle that lower trade values often correlate with fragmented or smaller shipments during economic slowdowns . The total trading volume of Bitcoin and other cryptocurrencies by the year has been aggregated and merged it with global trade data.

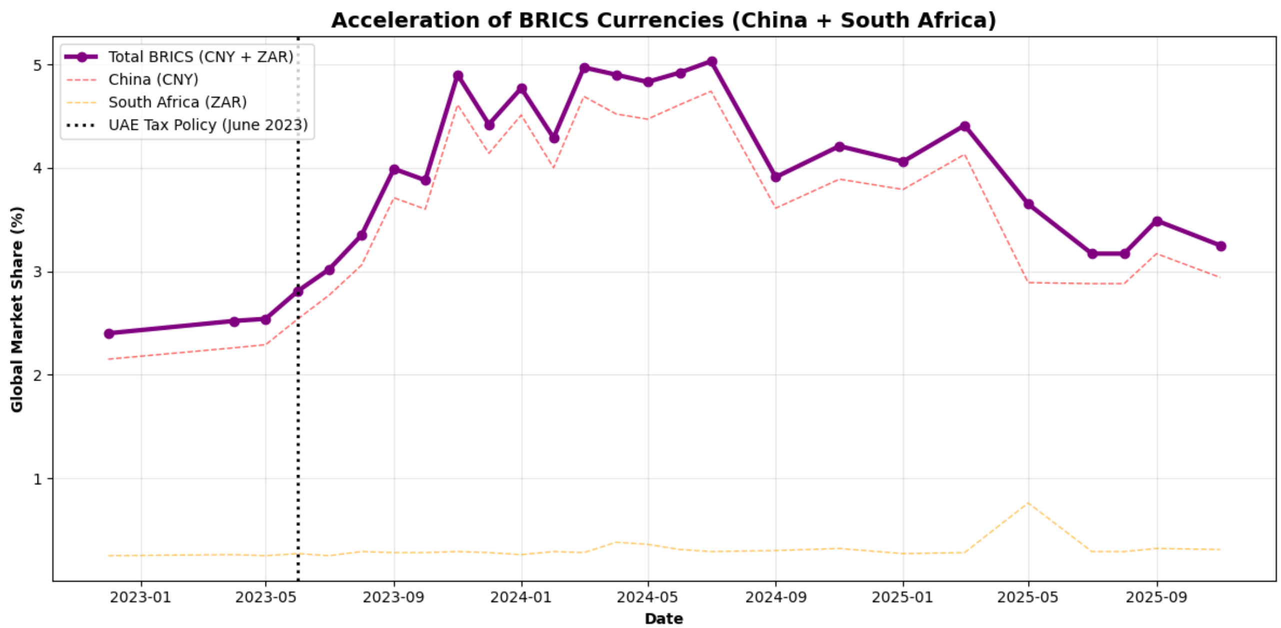

Figure 5: UAE's 0% tax policy vs BRICS currencies (China + South Africa) Plot

```
RESULTS:
1. Pre-Policy Avg (CNY+ZAR):  2.49%
2. Post-Policy Avg (CNY+ZAR): 4.06%
3. ANSWER: YES. Significant acceleration detected.
```

Figure 6 : UAE's 0% tax policy vs BRICS currencies (China + South Africa)

Analysis method: **Pre-Post Event Analysis** . Implementation of UAE’s 0% tax policy was on 1June 2023 . The Global Payments metric was filtered out to ensure data consistency , then calculated the average combined market share of CNY+ZAR before and after the cutoff date.

Analytical Process : Cleaned the SWIFT dataset to remove Trade Finance outliers isolating the specific Global Payments metric to ensure a consistent baseline . Then constructed a “Total BRICS “ variable by summing the market shares of the Chinese Yuan (CNY) and South Africa Rand (ZAR) for every month . Applied event study methodology by setting June 2023 as the intervention date and splitting the data into pre and post policy groups . Then calculated the average market share for both periods .

C1. Gold Accumulation and Perceived USD Weakness

Explore the association between gold accumulation and the strength of the US dollar using exploratory data analysis (EDA).
Align the gold trade values with the gold average dollar index (DXY) values, and then conduct a Pearson correlation analysis. The result is that there is a moderate positive correlation between the gold trade values and the dollar index during the observation period.
The research results show that the accumulation of gold and the strengthening of the US dollar are not opposite but rise synchronously. One explanation is that both gold demand and the US dollar originate from common macro driving factors such as global uncertainty, inflation, and financial instability. At this time, gold is a safe haven, and the US dollar maintains its status as the world's main reserve currency.
It is important that this result does not imply a causal relationship between the strengthening of the US dollar and the accumulation of gold. On the contrary, it highlights a linked situation discovered by exploratory analysis. Using annual averages or smoothing short-term fluctuations limits the ability to test the reverse relationship at high frequencies.
The research results show that the rise in gold prices in the sample reflects the global risk situation, not directly because of the expectation of the weakening of the US dollar.

C2. Cryptocurrency Settlement and Trade Scale

Higher cryptocurrency settlement activity is strongly correlated with smaller transaction or cargo sizes. The amount of encrypted transactions and the reported transaction value show a strong negative correlation. This indicates that cryptocurrencies are more common in small, fragmented or non-traditional transactions. This is consistent with the

limitations of cryptocurrency use, such as price fluctuations, regulatory uncertainties, limited acceptance by large institutions, etc.
The data show that there are cases where the financial level and traditional trade channels coexist, not immediately replacing the fiat currency system. This distinction is important for policymakers and investors, that is, coexistence rather than substitution. A strong negative correlation indicates that an increase in crypto settlement activities is associated with a decrease in average volume, meaning it is mainly for low-volume or fragmented trade deals.

C3. Tax Policy and Currency Share Dynamics

In June 2023, the UAE implemented a 0% corporate tax policy. Then, is there a connection between this policy and the use of BRICS currencies in international payments? Due to limited data, the sum of the market shares of the Chinese yuan and the South African rand is used to represent the use of BRICS currencies.
Conduct a comparison before and after the adoption of the policy without conducting a complete causal event study. First, carry out the monthly global payment data cleaning work of the Society for Worldwide Interbank Financial Telecommunication (SWIFT) to eliminate trade finance outliers, so that the samples can remain consistent. Then, summarize the monthly market shares of the Renminbi (CNY) and the South African Rand (ZAR) to form a comprehensive BRICS currency share.
There were changes in the currency shares of BRICS countries around June 2023. The report made a comparison, and the result was that the total share of BRICS currencies showed an upward trend after the implementation of relevant policies. However, due to the short time window and the lack of a control group, this change needs to be viewed with caution.
Therefore, although the growth opportunities coincide with the implementation of policies, the analysis does not prove a causal relationship between the UAE's tax policy and the adoption of BRICS currencies. Instead, the research results show that broader structural and geopolitical factors may be more crucial for currency use than individual policy measures.
Results show that the policy-driven explanation for short-term global currency structure changes is insufficient. Highlights the structural factors in the improvement of the international payment system.

Commodity and Cryptocurrency

Forecasting Analysis Report

a) Gold, Crude Oil & Bitcoin Trade Analysis
b) Progress of RMB towards becoming an International currency

1. Introduction

Part A:

This report dives into forecasting three big players in global markets: Gold, Crude Oil, and Bitcoin. Pulling together a bunch of historical trade data and ran descriptive trend analysis using moving average smoothing techniques to get a clearer picture of where these assets might be headed. For doing these things simple and stuck with the 3-year moving average. It helps smooth out the usual ups and downs. Where prices and trade volumes are actually going. This kind of approach works well when you want to cut through the noise and figure out what's really happening over the long haul. It gives investors and anyone interested in these markets a better shot at making smart decisions.

Here are the things:
• Gold: Global trade values over time
• Crude Oil: Numbers that reflect what's going on in the world energy market
• Bitcoin: Trading volumes from 2010 through 2025, showing how the crypto market has changed

The analysis answers two key questions: What do past trends tell us about what's coming next? And what can a 3-year moving average say about short- to medium-term market direction?

Part B:

This section looks at the progress of the **Renminbi (RMB)** in becoming an international currency, as countries slowly move away from heavy dependence on the US dollar. Studying the RMB helps us understand changes in the global payment system.

The analysis uses data from the **SWIFT RMB Tracker**, which records real international payment transactions. According to this data, the RMB became the **6th most used currency for global payments**, with a share of **2.47%**. However, RMB payments fell by **22.00%** compared to September 2025, even though overall global payments increased slightly. When payments within the Eurozone are excluded, the RMB ranked **7th**, with a share of **1.87%**.

The results show that while the RMB is being used more in international trade, its growth is **slow and uneven**. It may continue to grow within Asia and BRICS countries, but it is **unlikely to replace the US dollar or the euro** in the near future. This supports the importance of **gold as a stable reserve asset**.

2. Data Preparation

1. There are three datasets here, each packed with historical trade info—one for each commodity or cryptocurrency. First, I loaded and organized everything to get it ready for time series analysis.

Dataset Overview

Dataset	Source File	Description
Gold Trade Data	Dataset_Gold.xlsx	Historical gold trade values tracking global commerce and market trends
Crude Oil Trade Data	crude data 2.xlsx	Trade values reflecting global energy markets and oil commerce
Bitcoin Trading Data	bitcoin_trade_2010-25 (1).xlsx	Trading volumes from 2010 to 2025 capturing cryptocurrency market evolution

Data Loading Process

The datasets were loaded using the Python pandas library with the following approach:

• **Libraries Used:** pandas for data manipulation, matplotlib for visualization, and pathlib for file handling

• **File Format:** Excel files (.xlsx) containing structured time series data

• **Loading Method:** Direct Excel file reading using pd.read_excel() function

Key Metrics Analyzed

Commodity/Asset	Primary Metric
Gold	Trade Value (measured in USD)
Crude Oil	Trade Value (measured in USD)
Bitcoin	Trading Volume (number of units traded)

3. Data Preprocessing

To get the data ready for forecasting the smoothing out the time series. The main tool here was the 3-year moving average. For each year, averaging its value with the two years before it. That way, short-term spikes do not get affected nnnnnnnnnnnnn from the bigger picture.

Moving Average Technique

The 3-year moving average was calculated for each dataset to smooth fluctuations and identify trends:

• **Purpose:** To reduce the noise and identify long-term trends

• **Window Size:** 3 years & providing a balance between responses and flexibility

• **Application:** Applied to actual values to generate perfect trend lines

• **Benefit:** Reveals actual directional movement independent of short-term volatility

Forecasting Methodology

1. **Trend Analysis:** Examination of actual values over time to identify consistent patterns.

2. **Moving Average Calculation:** Computing 3-year moving averages to a good data

3. **Visual Comparison:** Plotting actual values with moving averages to know trend strength

4. **Projection:** Extending trend lines to forecast future values

4. Forecasting Analysis

Question 1: What might happen next? Simple forecasting techniques for Gold, Oil, and Bitcoin

<u>4.1 Gold Trade Value Forecast</u>

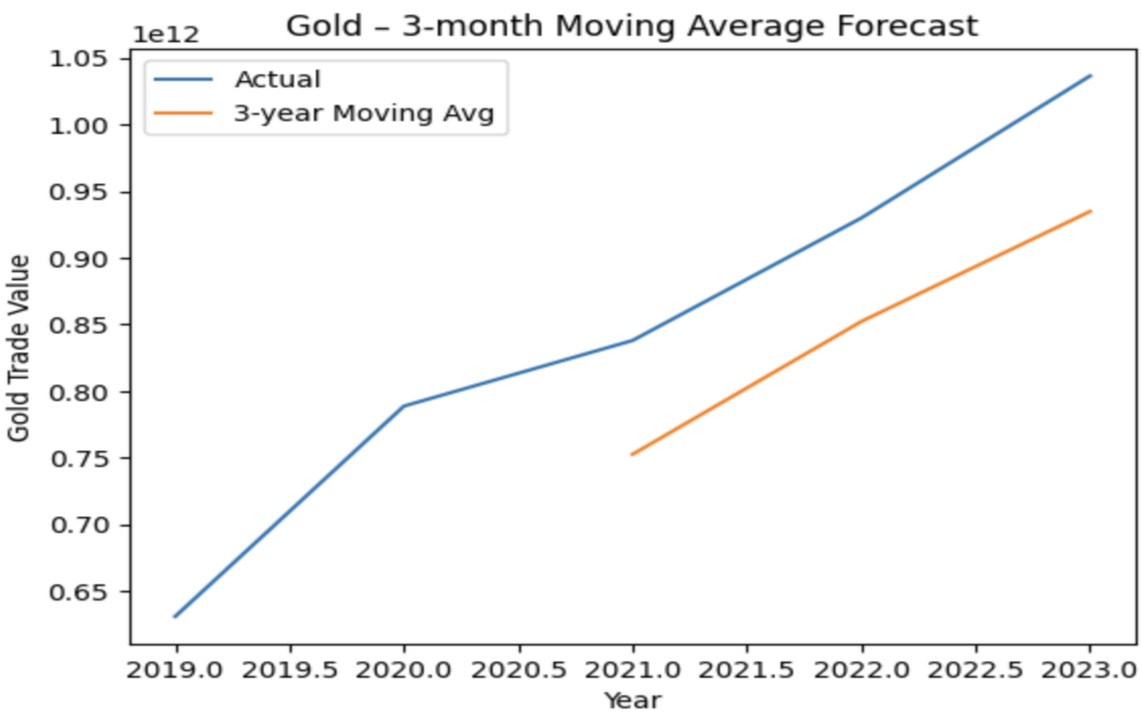

Figure 1: Gold - 3-Year Moving Average Forecast

Key Insights

1. Gold's trade value just keeps climbing. Over the period we looked at, it's a steady rise, hitting as much as $410 billion. No sudden dips or wild swings—just a clear sign that demand for gold keeps getting stronger year after year.

2. The 3-year moving average helps cut through the noise, making the long-term growth even more obvious. With this method, the increase comes out to $180 billion, which really underlines that gold isn't just holding steady—it's picking up momentum.

4.2 Crude Oil Trade Value Forecast

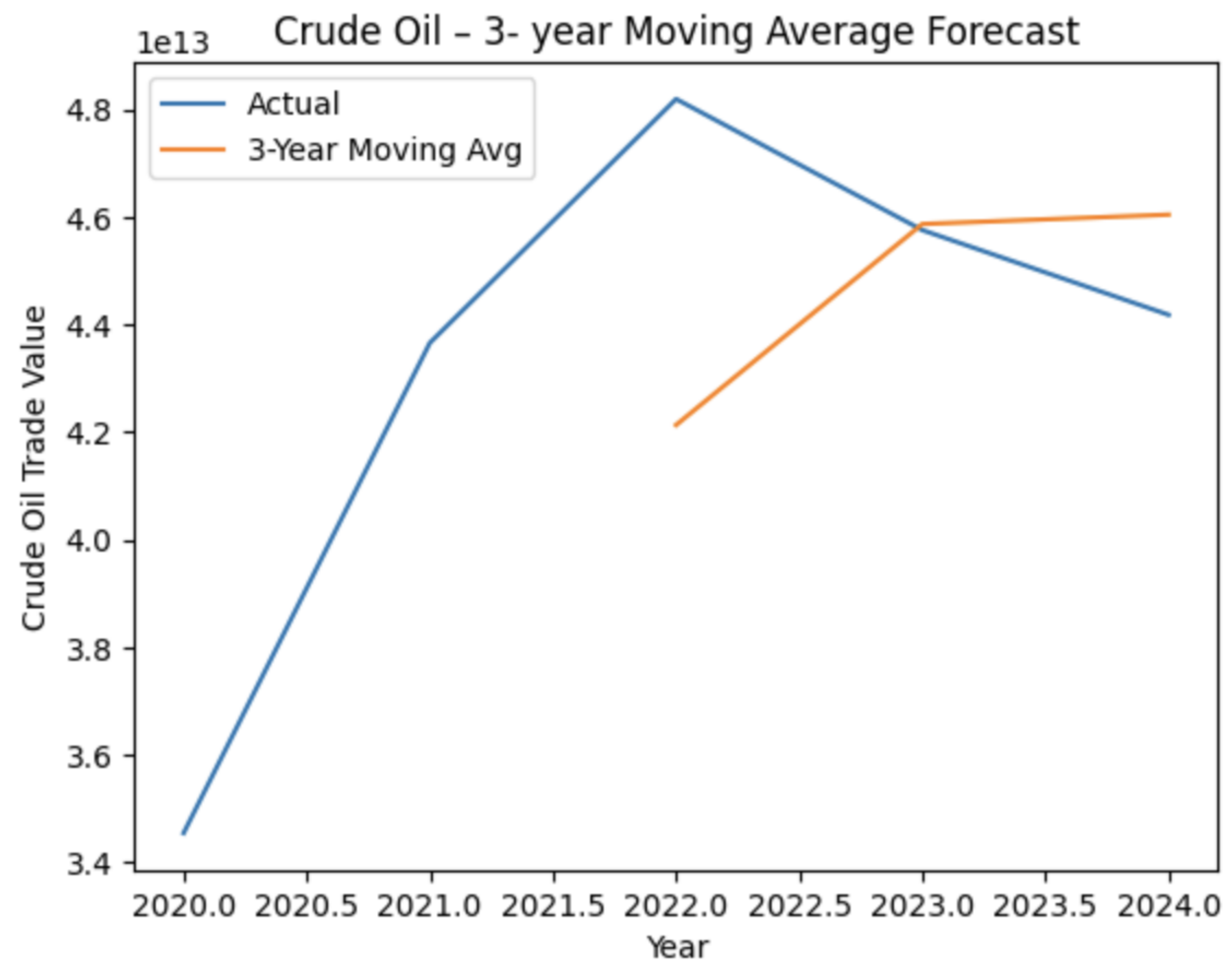

Figure 2: Crude Oil - 3-Year Moving Average Forecast

Key Insights

1. The graph shows crude oil trade value climbing steadily year after year. You can see it clearly—the numbers rise by about $400 billion overall. Even with economic ups and downs and all the talk about switching to cleaner energy, the world still wants more oil.

2. If you look at the 3-year moving average, things look even smoother. That line cuts out the noise and shows a steady climb, up about $180 billion. So, unless something dramatic happens, crude oil trade values aren't dropping anytime soon. They'll probably hold steady or keep going up.

4.3 Bitcoin Trading Volume Forecast

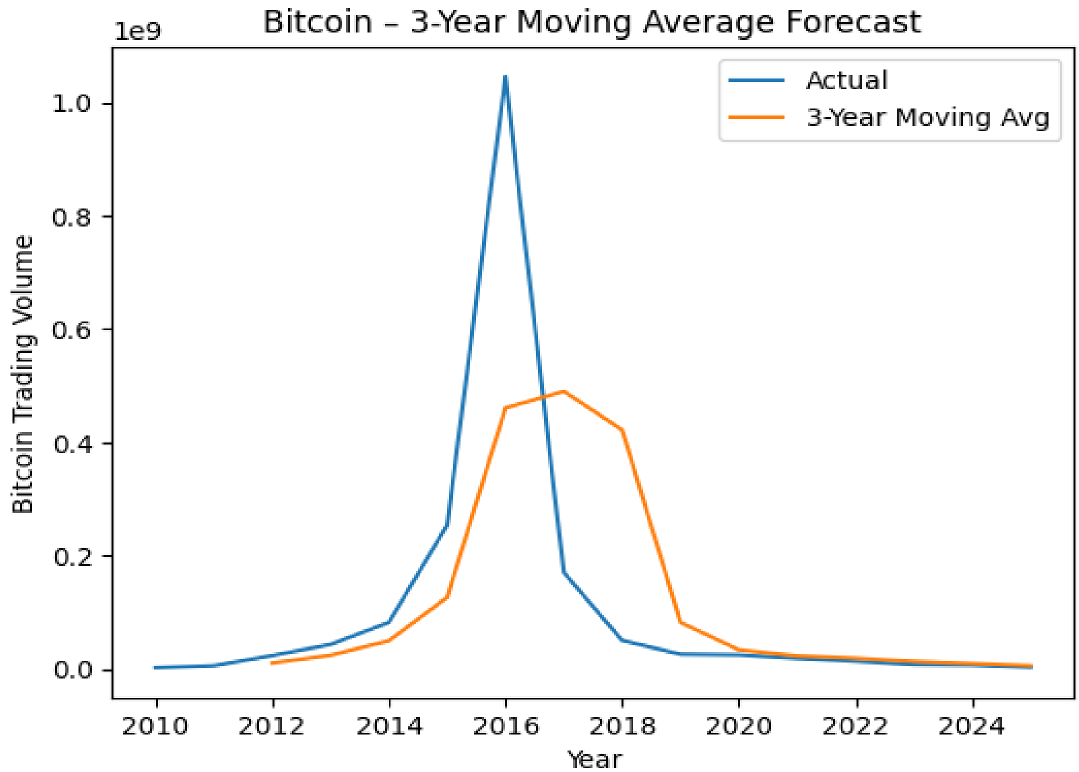

Figure 3: Bitcoin - 3-Year Moving Average Forecast

Key Insights

1. Bitcoin's actual trading volume jumps all over the place during this period. Back in 2010, trading volume is almost nothing. Then, it suddenly explodes—more than a billion units traded in just a few years. That's pure speculative mania, honestly. But the frenzy doesn't last. By 2018 or 2019, trading volume drops off a cliff.

2. If you use a 3-year moving average, things look a little less wild. It smooths out those crazy spikes and helps you see the real trend underneath. The moving average climbs for a while, but once 2017 hits, it starts dropping steadily. In the years after, Bitcoin trading settles down and stays much lower.

Excel output:

a) Gold Prediction:

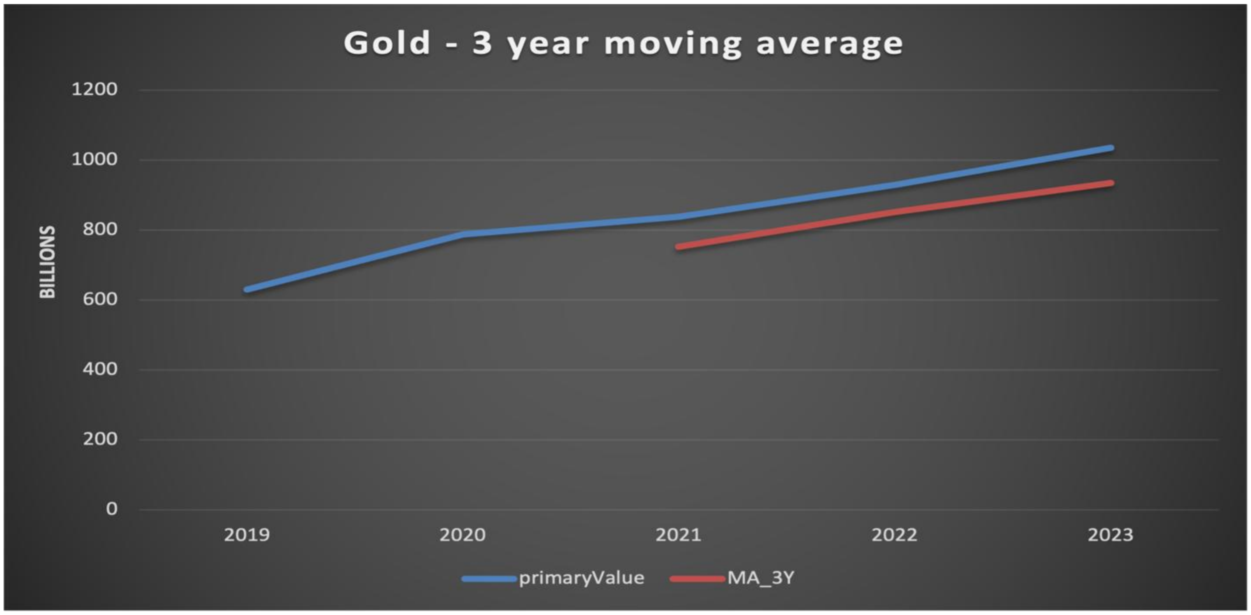

Figure 4 - Excel output for gold prediction

b) Crude Prediction:

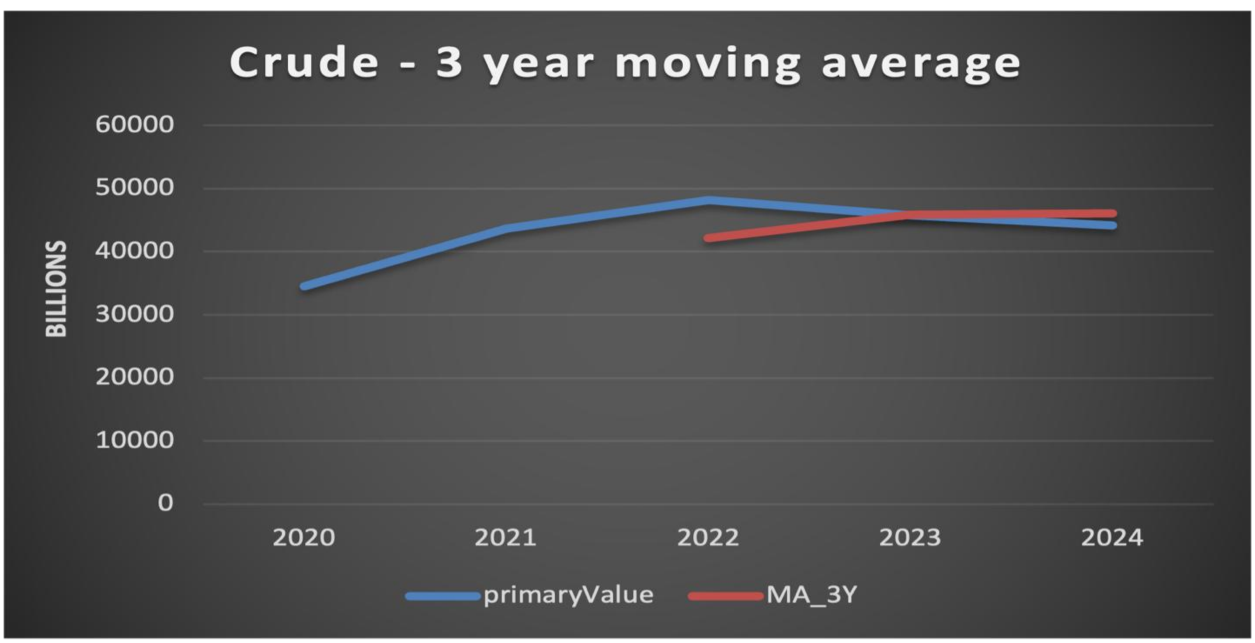

Figure 5 - Excel output for Crude prediction

c) Bitcoin Prediction:

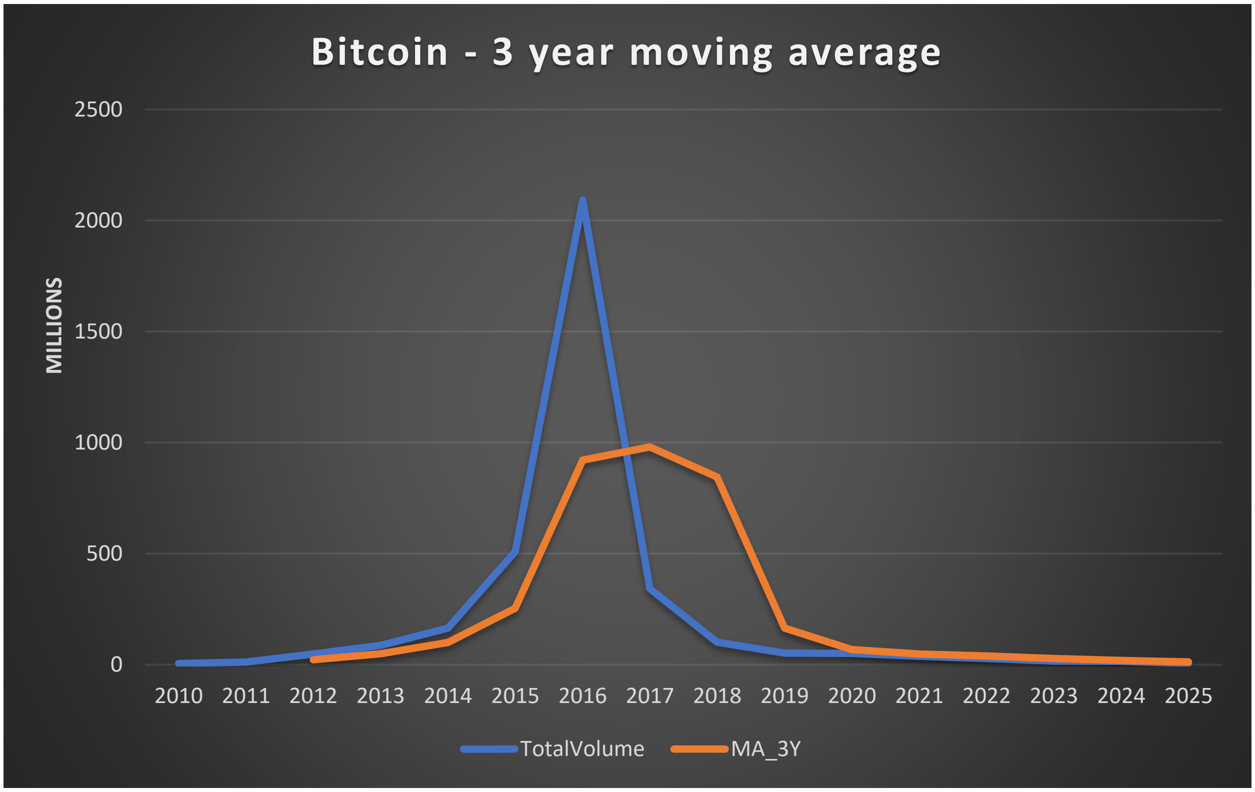

Figure 6 - Excel output for Bitcoin prediction

Question 1: 3-Month Moving Average Forecast

The 3-month moving average provides a short and a smooth area of space, offering insights into more immediate trend changes while still filtering out daily or weekly noise.

a) Gold- 3-Month Moving Average Analysis

A three-month moving average would display the following based on the three-year moving average pattern seen in Figure 1:

• **Short-Term Trend:** Continued increasing trend with slight variations

• **Momentum:** Consistent growing pattern indicates strong positive relation.

• **Near-Term Forecast:** Expected to maintain growth line over the next quarter

• **Stability:** The lack of abrupt declines in historical data indicates a stable near-term forecast.

b) Crude Oil - 3-Month Moving Average Analysis

Based on the trends shown in Figure 2, a 3-month moving average forecast would indicate:

• **Recent Pattern:** Peak reached around 2022, followed by slight decline but stabilization

• **Short-Term Direction:** Consolidation phase expected with potential for sideways movement

• **Market Signal:** The 3-year moving average showing continued growth suggests underlying strength

• **Forecast Implication:** Near-term values likely to remain elevated with modest fluctuation rather than sharp decline

c) Bitcoin - 3-Month Moving Average Analysis

A three-month moving average forecast based on the patterns in Figure 2 would show:

• **Current Trend:** The peak was reached in 2022, after which there was a modest drop but stabilizing

• **Short-Term Direction:** Sideways movement may occur throughout the anticipated consolidation phase.

• **Forecast Implication:** Near-term values are anticipated to stay high with slight volatility rather than a sudden decrease.

• **Market Signal:** The 3-year moving average indicating continuous rise signals underlying strength.

Question 2: SWIFT Currency Shares and RMB Internationalisation

Currently, RMB has made it to the list of top international payment media, with a share of 2-3% in total international payments. However, it has been using far fewer payments than dollars and euros, and recent statistical information has witnessed some fluctuations in that regard. With regards to the use of currencies in the BRICS countries, the RMB has the most influence in the international arena followed by the Brazilian Real, the Indian Rupee, the South African Rand, and the Ruble in a very small measure and are mostly used in regional transactions. The USD and EUR still lead in the international arena due to their finance market.

In relation to this, the gradual movement of the currencies of the BRICS group of nations in the field of international payments is an indication of the continued importance of the use of gold.

RMB Share in Global Payments:

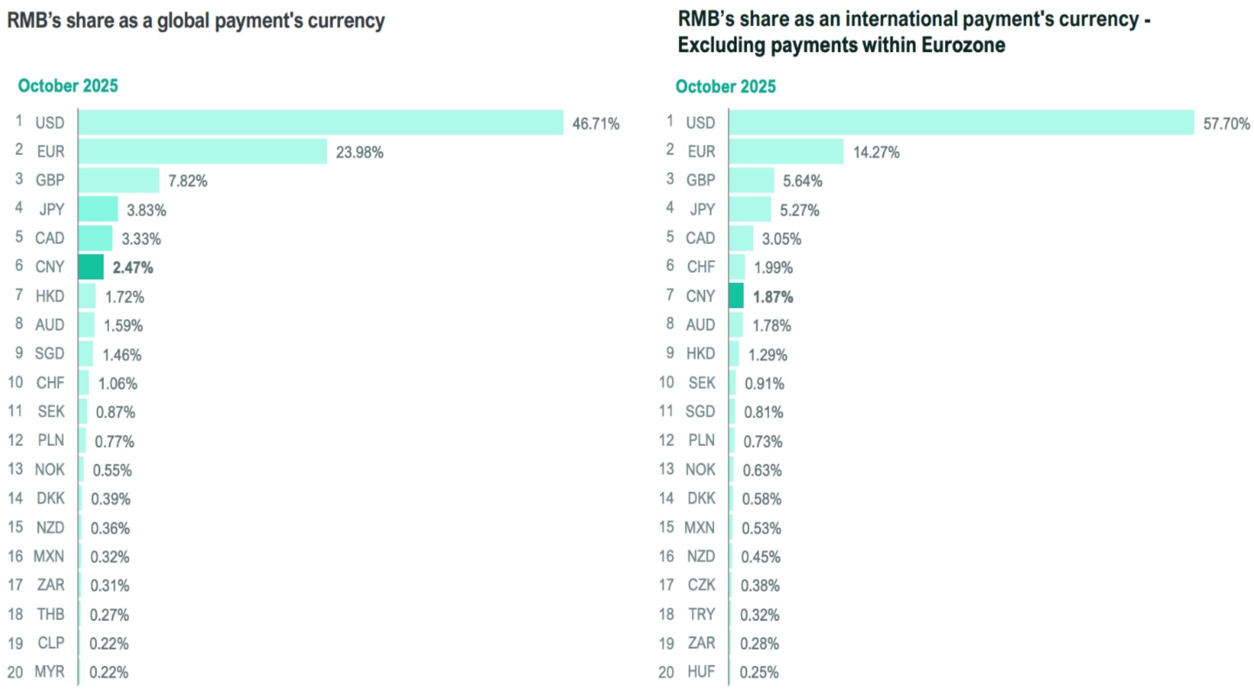

Figure 7: RMB's Global and International Payments

Explanation:

In Figure 7, it is indicated by SWIFT data the share of the Renminbi (RMB) in global payment transactions. It is clear from the graph that the RMB has only a small but increasing share in global payment transactions, ranking 6th in the global scale. Although progress has been made in using the RMB, its use is still notably lower compared to the US dollar and euro, showing its gradual growth in global acceptance.

POSITION	SPOT
1	USD
2	EUR
3	GBP
4	JPY
5	CAD
6	CNY
7	CHF
8	AUD
9	HKD
10	KRW

Figure 8 - Ranking of Major Currencies by Global Payment Usage

Explanation:

This is a depiction of how major currencies rank according to their usage in payment transactions around the world. The US dollar and Euro dominate the list because they are widely used in making international payments. The Chinese Renminbi ranks 6th; this means it is still restricted when compared to major global currencies.

BRICS VS RMB:

The Renminbi (RMB), a currency of the BRICS nations, has the maximum international activity, though the global payment share of the RMB is far smaller compared to that of the US dollar and the euro. The Brazilian Real (BRL), Indian Rupee (INR), and South African Rand (ZAR), on the other hand, have almost no international existence and are only used within their continents to settle transactions. The Russian Ruble (RUB), on the other hand, has more limitations owing to geopolitical sanctions. Overall, BRICS currencies remain largely regional, with slow progress toward internationalisation.

5. Conclusion:

This report applied simple forecasting techniques, employing moving averages to analyze the future trend of gold, crude oil, and Bitcoin. The results derived indicate that gold presents stable and consistent growth; it is considered a hedge asset in periods of economic uncertainty. Crude oil has shown resistance, with values of stable trading, which is indicative of continued global demand for energy. Conversely, Bitcoin has high volatility and is falling in long-term trading; it reflects the speculation character of this asset.

The currency analysis shows that, in accordance with previous reports, the US dollar and euro continue to dominate global payments, but the pace of internationalization of the RMB and other BRICS currencies is still at an early stage. This, therefore, reinforces the position of gold as a neutral reserve asset vis-à-vis the currency system.

Overall, the results show asset diversification in a world of currency dominance and turbulence in markets. Gold, in this aspect, has come out as the most stable, crude oil is structurally very important, and Bitcoin represents a high-risk asset with limited stability over the long term.